

Token Limits and Context Windows in Large Language Models

1. What Are Tokens?

Tokens are the basic units of text processed by a Large Language Model (LLM). A token may represent a whole word, part of a word, punctuation, or a special symbol. Before processing, the input text is converted into tokens using a tokenizer. Each token is then transformed into a numerical representation (embedding) that the model can process.

2. What Is a Context Window?

A context window is the maximum number of tokens an LLM can consider at one time while generating a response. It determines how much of the input the model can remember and use as context. Larger context windows allow the model to process longer documents and maintain consistency across more text.

3. What Happens When the Input Exceeds the Token Limit?

If a legal document contains more tokens than the model's context window can handle, the model cannot process the entire document at once. Depending on the system, extra tokens may be truncated or omitted, causing important clauses, evidence, or conclusions to be ignored. As a result, summaries may become incomplete or inaccurate.

4. Two Practical Solutions for Handling Long Inputs

Solution 1: Chunking

Split the document into smaller sections that fit within the context window. Summarize each section separately and then combine the summaries into a final summary.

Solution 2: Retrieval-Augmented Generation (RAG)

Store document sections in a searchable database and retrieve only the most relevant sections for the user's query before sending them to the LLM.

5. Real-World Impact on Legal AI Systems

Legal documents often contain critical information spread across many pages. If important sections are omitted because of token limits, the AI may miss legal obligations, exceptions, deadlines, or evidence. This can lead to incorrect legal summaries and poor decision support. Using techniques such as chunking, retrieval, and human review improves accuracy, reliability, and trust in legal AI applications.